Complex Tori and Elliptic Curves

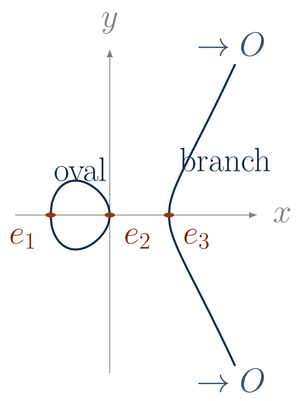
Companion Gallery — six further pictures of the same object

Read alongside the main poster: the real and finite faces of E, the torsion and moduli structure, and the degenerations

Gallery — More Pictures of the Same Story

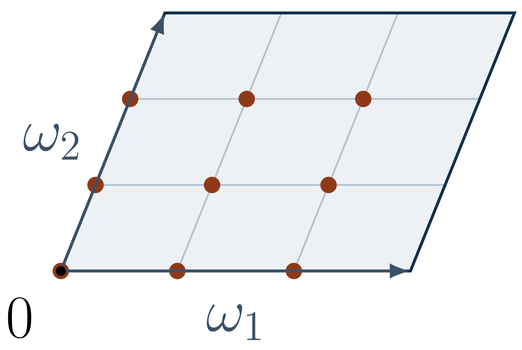
The main poster carries the argument that the chord–tangent law on a cubic *is* addition on the torus $\mathbb{C}/\Lambda$. This companion sheet enriches it with six further pictures of the *same* object: the real and finite-field faces of E, the second Weierstrass function $\wp'$, the torsion and moduli structure, and the degenerations that force the discriminant $\Delta \neq 0$.

The real picture: two components



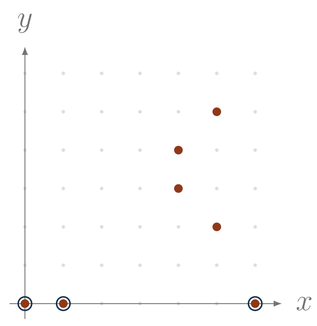
Over $\mathbb{R}$, $E: y^2 = 4x^3 - 4x$ has a compact oval *over $[e_1, e_2]$ and an* unbounded branch *over $[e_3, \infty)$, both ends running to O at infinity. The three roots together with O are the 2-torsion $E[2] = \{O, e_1, e_2, e_3\} \cong (\mathbb{Z}/2\mathbb{Z})^2$.*

Torsion: $E[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$



The n-torsion is $\frac{1}{n}\Lambda/\Lambda$ — a regular $n \times n$ grid inside one tile, so $E[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$. Shown for $n = 3$: the nine three-torsion points, exactly the flexes of E.

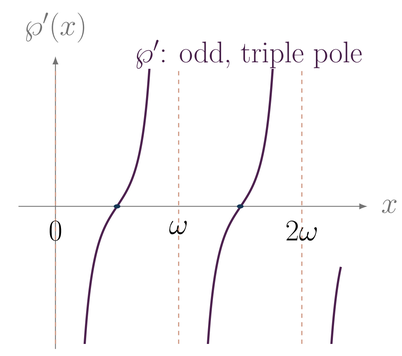
Over a finite field: $E(\mathbb{F}_7)$



$\#E(\mathbb{F}_7) = 7 + 1 = 8$ (circled: 2-torsion)

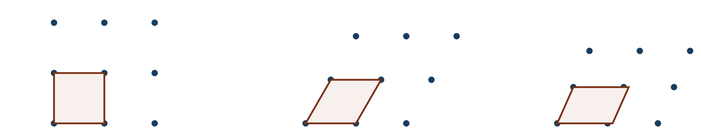
Because the addition law is rational, *it works mod p. Solving $y^2 = 4x^3 - 4x$ over $\mathbb{F}_7$ gives seven affine points; with O, $\#E(\mathbb{F}_7) = 8$. The full 2-torsion (circled) makes $E(\mathbb{F}_7) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/4$.*

The other function: $\wp'$



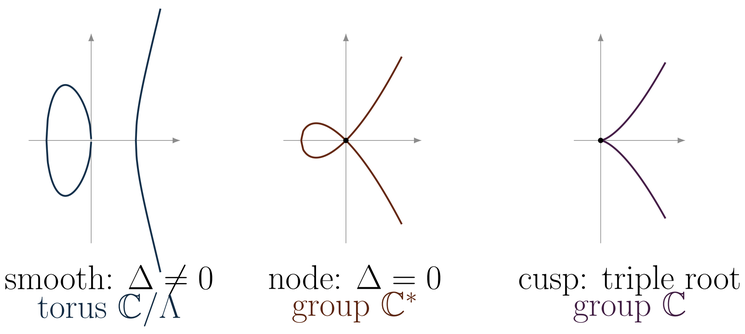
Beside $\wp$ lives its derivative $\wp'$: odd, with a triple *pole at every lattice point, vanishing at the half-periods (where $\wp$ has its extrema). The pair $(\wp, \wp')$ is precisely the (x, y) of the embedding $\Phi = [\wp : \wp' : 1]$.*

Moduli: shape sets the j-invariant



Homothetic lattices give the same torus; the shape $\tau \in \mathbb{H}$, taken modulo $\mathrm{SL}_2(\mathbb{Z})$, is recorded by the j-invariant. The square ($\tau = i$, $j = 1728$) and hexagonal ($\tau = \rho$, $j = 0$) lattices are the two extra-symmetric ones.

Why $\Delta \neq 0$: node and cusp



As $\Delta \to 0$ the torus degenerates: two roots colliding give a node *(group $\mathbb{C}^*$), and a triple root a* cusp *(group $\mathbb{C}$). Only the smooth case, $\Delta \neq 0$, is an elliptic curve.*

LaTeX TikZposter